SDC Simulation Experiment

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Abstract

In this paper I present a simple approach to running stochastic simulations of the SDC process - specifically to demonstrate the parity process

Keywords: simulations, dna computing

1 Introduction

Further to the research I have conducted within the field of DNA computing [1], I have implemented a simple, yet extensible, framework for creating stochastic, or Monte Carlo, simulations to gain qualitative insights into the DNA computing process.

2 The Process

The key points of the simulation process can be described as follows.

1. for each scaffold, randomly pick a tile from the solution
   - if the tile position is vacant, place it and proceed
   - replace (probability) the existing tile if the bind is imperfect
   - otherwise continue from the beginning with the next scaffold
2. continue until the experiment concludes
3. collect completed scaffolds, and count successes
4. report on the outcome

3 The Implementation

The simulation was implemented in the Python programming language. Executing the simulation looks as follows:

```
sdc  = SDCParitySimulation(...)
data = sdc.run(iterations)
```

The parameters of the experiment are passed to the simulation object through the constructor, and the number of iterations are specified when running the simulation.

4 The Results

Currently the simulation collects success rates over the iterations, which are printed at the conclusion of the experiment. These collected success rates may be used for plots and further statistical analysis.

```
    A Simulation of Scaffolded DNA Computer: Parity


Results after 10 iterations

 Iteration: 1
     Count: 6264
Proportion: 62.64%

 Iteration: 2
     Count: 6428
Proportion: 64.28%

 Iteration: 3
     Count: 6137
Proportion: 61.37%

...
```

5 Conclusion

Simulations such as this could form the basis of a qualitative analysis of the SDC process if enhanced to include other parameters such as temperature, energy, and more.

References

[1] Tristan Stérin, Abeer Eshra, Janet Adio, Constantine Glen Evans, and Damien Woods. A thermodynamically favoured molecular computer: Robust, fast, renewable, scalable. *bioRxiv*, 2025.